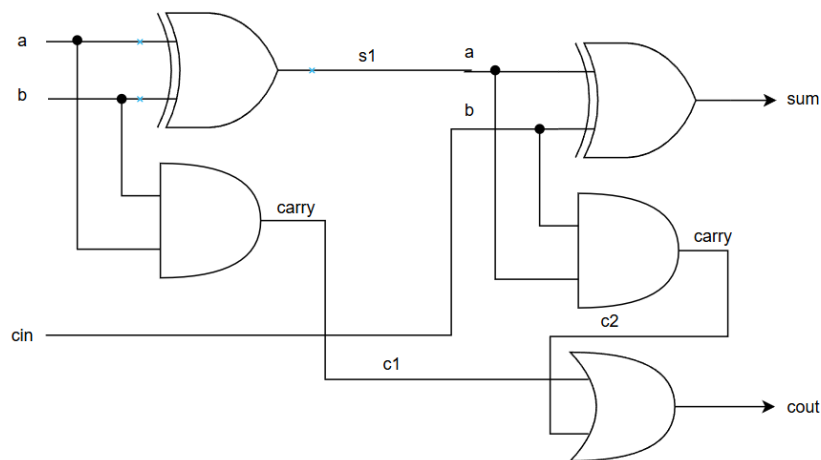


Task 3

<https://edaplayground.com/x/bkgc>

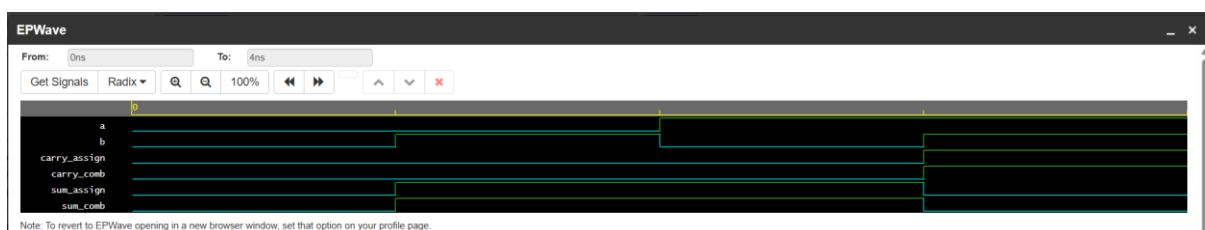


Full Adder by using 2 Half Adders

Gate Level Diagram

```
# ELAB2: Elaboration final pass complete - time: 0.4 [s].
# KERNEL: SLP loading done - time: 0.0 [s].
# KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.
# KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.
# KERNEL: SLP simulation initialization done - time: 0.0 [s].
# KERNEL: Kernel process initialization done.
# Allocation: Simulator allocated 4540 kB (elbread=427 elab2=4106 kernel=6 sdf=0)
# KERNEL: ASDB file was created in location /home/runner/dataset.asdb
# KERNEL: a b | assign(sum,carry) | comb(sum,carry)
# KERNEL: 0 0 | 0 0 | 0 0
# KERNEL: PASS
# KERNEL: 0 1 | 1 0 | 1 0
# KERNEL: PASS
# KERNEL: 1 0 | 1 0 | 1 0
# KERNEL: PASS
# KERNEL: 1 1 | 0 1 | 0 1
# KERNEL: PASS
# RUNTIME: Info: RUNTIME_0068 testbench.sv (43): $finish called.
# KERNEL: Time: 4 ns, Iteration: 0, Instance: /tb, Process: @INITIAL#24_0@.
# KERNEL: stopped at time: 4 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
# VSIM: Simulation has finished.
Done
```

Terminal Output Screenshot



Waveform Screenshot

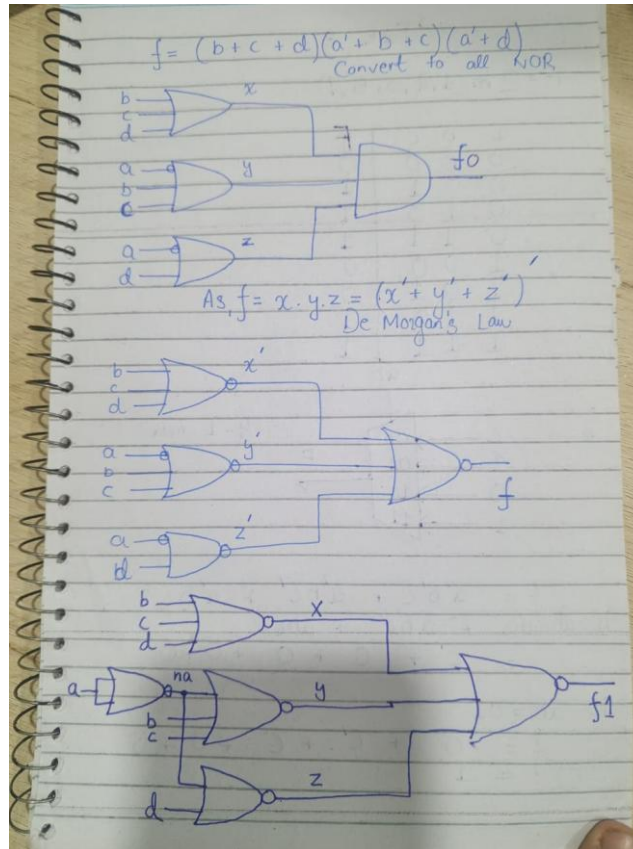
During synthesis, the synthesis tool analyzes the logic equations rather than the coding style.

It recognizes that both modules implement identical combinational logic and therefore generates the same hardware gates (one XOR gate for the sum and one AND gate for the

carry). Thus, different RTL coding styles can synthesize to identical circuits as long as they describe the same combinational behaviour.

Task 4

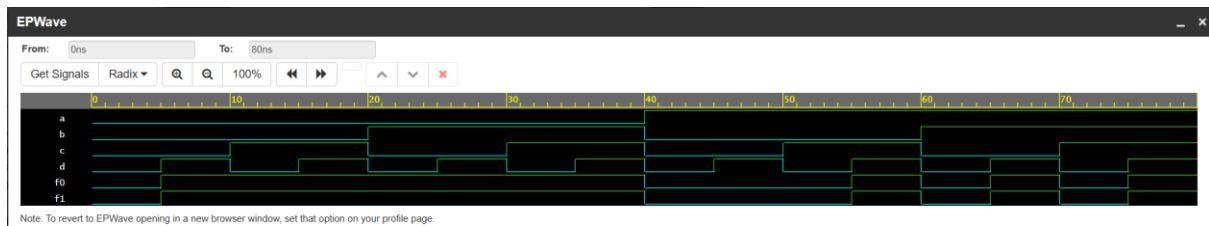
<https://edaplayground.com/x/6pxX>



Gate Level Diagram

```
# KERNEL: PASS! for abcd: 0000
# KERNEL: PASS! for abcd: 0001
# KERNEL: PASS! for abcd: 0010
# KERNEL: PASS! for abcd: 0011
# KERNEL: PASS! for abcd: 0100
# KERNEL: PASS! for abcd: 0101
# KERNEL: PASS! for abcd: 0110
# KERNEL: PASS! for abcd: 0111
# KERNEL: PASS! for abcd: 1000
# KERNEL: PASS! for abcd: 1001
# KERNEL: PASS! for abcd: 1010
# KERNEL: PASS! for abcd: 1011
# KERNEL: PASS! for abcd: 1100
# KERNEL: PASS! for abcd: 1101
# KERNEL: PASS! for abcd: 1110
# KERNEL: PASS! for abcd: 1111
# KERNEL: Simulation has finished. There are no more test vectors to simulate.
# VSIM: Simulation has finished.
Done
```

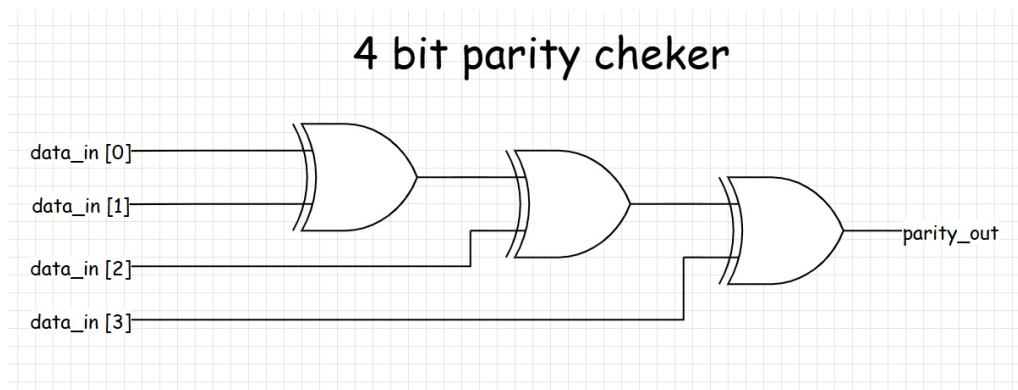
Terminal Output Screenshot



Waveform Screenshot

Task 5

<https://edaplayground.com/x/FTQh>

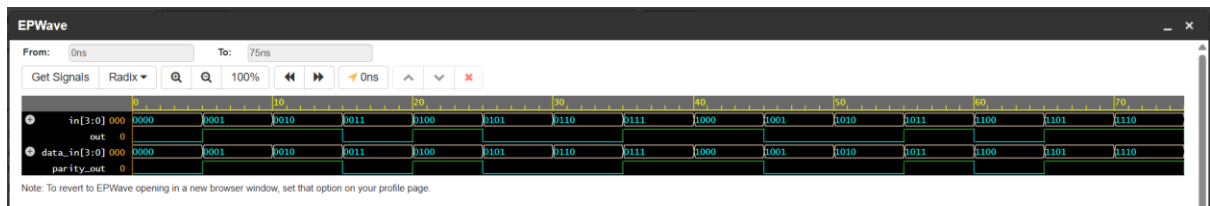


Gate Level Diagram

```

Log Share
CPU time: .605 seconds to compile + .503 seconds to elab + .544 seconds to link
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 20 04:17 2026
PASS! for in: 0000
PASS! for in: 0001
PASS! for in: 0010
PASS! for in: 0011
PASS! for in: 0100
PASS! for in: 0101
PASS! for in: 0110
PASS! for in: 0111
PASS! for in: 1000
PASS! for in: 1001
PASS! for in: 1010
PASS! for in: 1011
PASS! for in: 1100
PASS! for in: 1101
PASS! for in: 1110
PASS! for in: 1111
VCS Simulation Report
Time: 80 ns
CPU Time: 0.540 seconds; Data structure size: 0.0Mb
Mon Jul 20 04:17:39 2026
Done
  
```

Terminal Output Screenshot



Waveform Screenshot